Parity and binary divisibility for compositional power series

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Contribution and responsibility. This report presents mathematical arguments and a Lean formalization developed autonomously by ChatGPT (OpenAI) from prompts by Dev Goyal. Dev initiated the investigation and requested checks and revisions. He did not supply the proof strategy or intermediate mathematical arguments. Dev has confirmed that human verification is complete. He is the sole human author and accepts responsibility for the report. The series recorded by Paul D. Hanna and Danial Asaria's prior counterexample receive credit below.

Abstract

This report studies the integer coefficients of the formal power series $A_m(x) = x + \sum_{n\geq 2} a_n(m)x^n$ determined by $A_m(xA_m(x) - mxA_m(x)^2) = x^2$. We prove that every $a_n(m)$ with $n \geq 1$ is odd when m is odd. This proves the parity assertion recorded for $m = 3$ in OEIS A396843. For each $m \geq 2$, we determine every coefficient modulo m^2. We also prove that $m^{s_2(n-1)}$ divides $a_n(m)$, where $s_2(r)$ is the number of ones in the binary expansion of r. This divisibility bound also holds for the coefficients of the compositional inverse of A_m. Combining the two divisibility results identifies exactly the indices at which a prime divisor of m occurs to its smallest possible exponent. The proofs use coefficient comparison, modular arithmetic, and induction; the formal power series facts needed in the argument are established explicitly.

1 Introduction

A generating function records a sequence as the coefficients of a power series. An equation for that series can impose arithmetic patterns that are not apparent from its first few terms. We consider the family

$$A_m(xA_m(x) - mxA_m(x)^2) = x^2, \qquad A_m(x) = \sum_{n\geq 1} a_n(m)x^n, \qquad a_1(m) = 1, \tag{1}$$

where m is an integer. The constant coefficient is zero. The expression $A_m(U(x))$ means substitution of $U(x)$ for x in A_m, whereas $A_m(x)^2$ means ordinary multiplication of two copies of the series.

Instances of this family were recorded by Paul D. Hanna in the On-Line Encyclopedia of Integer Sequences (OEIS). The parameters $m = 1, 2, 3, 4, 5$ correspond respectively to A265940, A271958, A396843, A396844, and A396845 [1, 2, 3, 4, 5]. For $m = 3$, the series begins

$$A_3(x) = x + 3x^2 + 15x^3 + 117x^4 + 975x^5 + 8631x^6 + \cdots .$$

The entry A396843 conjectures that all its nonconstant coefficients are odd [3]. We prove this for every odd m.

The divisibility results have a different pattern. For $m \geq 2$, every coefficient after the first is divisible by m, and its remainder modulo m^2 is nonzero exactly when its index is one more than a power of two. A stronger bound depends on the whole binary expansion of the index:

$$m^{s_2(n-1)} \mid a_n(m).$$

Here $d \mid q$ means that q is an integer multiple of d, and $s_2(r)$ counts the ones in the binary expansion of the nonnegative integer r. For example, $31 = (11111)_2$, so the bound gives $3^5 \mid a_{32}(3)$. The bound is a guaranteed divisor, not a claim that no larger power divides the coefficient.

The argument proceeds in three stages. First, a change in one coefficient shows that (1) determines a unique series. This lets us prove parity by exhibiting its reduction modulo two. Second, an expansion modulo m^2 gives the exact residue formula. Finally, we use the series and its compositional inverse together in an induction based on binary digits.

2 Formal power series and notation

2.1 Operations and coefficient comparison

A formal power series is an expression $U(x) = \sum_{n \geq 0} u_n x^n$ whose coefficients are considered algebraically. We do not substitute numerical values of x, and no convergence assumption is needed. Equality means equality of every coefficient. We write $[x^n]U = u_n$. Addition is coefficientwise, and multiplication is defined by

$$[x^n](UV) = \sum_{i=0}^{n} u_i v_{n-i}.$$

Thus each coefficient of a product is a finite sum. An empty sum is zero.

If $V(0) = 0$, composition is defined by

$$U(V(x)) = \sum_{n \geq 0} u_n V(x)^n.$$

The power $V(x)^n$ has no terms of degree below n, so each coefficient again receives only finitely many contributions. For example, if $U(x) = x + x^2$ and $V(x) = x^2$, then $U(V(x)) = x^2 + x^4$. Composition of series with zero constant coefficient is associative: one can verify this in each degree by the usual rules for polynomials, since only finitely many terms contribute.

The notation $O(x^d)$ denotes an unspecified series whose coefficients below degree d vanish. In particular, $U = x + O(x^2)$ means that U has constant coefficient zero and coefficient of x equal to one. We call this *normalized*. Products and compositions satisfy

$$O(x^a)O(x^b) = O(x^{a+b}), \qquad U(V(x)) = O(x^{ab}) \quad \text{if } U = O(x^a),\ V = O(x^b),\ b \geq 1.$$

These statements specify vanishing of coefficients, not estimates as a numerical variable tends to zero.

We occasionally allow coefficients in a commutative ring R with identity: a coefficient system in which addition, subtraction, and multiplication obey their usual algebraic laws, with a multiplicative identity 1. The examples used here are the integers $\mathbb{Z}$, residues $\mathbb{Z}/q\mathbb{Z}$, and integer polynomials $\mathbb{Z}[t]$. The notation $R[[x]]$ denotes formal power series with coefficients in R.

For integer series, $U \equiv V \pmod{q}$ means that every coefficient of $U - V$ is divisible by q. Reducing coefficients modulo q commutes with addition, multiplication, and every composition

used here: each coefficient is obtained by a finite sequence of integer operations. In particular, $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ has the two elements 0 and 1, and addition equals subtraction in this ring.

If $U(0) = 0$, the series $1 + U$ has a multiplicative inverse, given formally by

$$(1+U)^{-1} = \sum_{k\geq 0} (-U)^k.$$

The expression is well defined coefficientwise, and multiplication makes successive terms cancel. This justifies fractions such as $x/(1+x)$.

2.2 Compositional inverses

A compositional inverse reverses substitution rather than multiplication. We will need both identities in the following elementary fact.

Lemma 2.1. *For any commutative ring R with identity and any normalized series $A \in R[[x]]$, there is a unique normalized series $B \in R[[x]]$ such that*

$$A(B(x)) = B(A(x)) = x.$$

Proof. Write $A = x + \sum_{n\geq 2} a_n x^n$ and seek $B = x + \sum_{n\geq 2} b_n x^n$ with $A(B) = x$. In degree $n \geq 2$, the linear term B contributes b_n. In every higher power B^k, with $k \geq 2$, a contribution to degree n uses only coefficients of B of index at most $n - k + 1 < n$. Consequently, the coefficient equation in degree n determines b_n uniquely from earlier coefficients, using only addition, subtraction, and multiplication. This constructs the unique normalized B with $A \circ B = x$.

Apply the same construction to B to obtain a normalized C with $B \circ C = x$. Associativity gives

$$A = A \circ (B \circ C) = (A \circ B) \circ C = C.$$

Hence $B \circ A = B \circ C = x$ as well. Uniqueness follows from the construction of the right inverse. □

3 Existence and uniqueness of the defining series

The following coefficient comparison is the reason modular solutions can be identified by direct substitution.

Lemma 3.1. *Let R be a commutative ring with identity and let $m \in R$. There is exactly one normalized series $A \in R[[x]]$ satisfying*

$$A(xA - mxA^2) = x^2.$$

Proof. For a normalized series A, set $T_A = xA - mxA^2$ and $\Psi_m(A) = A(T_A)$. We first show how a change in one coefficient affects the defining equation:

$$B - A = cx^r + O(x^{r+1}), \quad r \geq 2 \quad \Longrightarrow \quad \Psi_m(B) - \Psi_m(A) = cx^{r+1} + O(x^{r+2}). \tag{2}$$

Indeed,

$$T_B - T_A = x(B-A)(1 - m(A+B)) = cx^{r+1} + O(x^{r+2}).$$

Both T_A and T_B start with x^2. If $A = x + \sum_{k\geq 2} a_k x^k$, then

$$A(T_B) - A(T_A) = (T_B - T_A) + \sum_{k\geq 2} a_k (T_B^k - T_A^k).$$

For $k \geq 2$, factor $T_B^k - T_A^k$ as $T_B - T_A$ times a sum of products of $k-1$ inner arguments. Its degree is therefore at least $r+1+2(k-1) \geq r+3$. It follows that

$$A(T_B) - A(T_A) = (T_B - T_A) + O(x^{r+3}).$$

The other difference, $(B-A)(T_B)$, is $O(x^{2r})$, hence $O(x^{r+2})$. Adding the two differences proves (2).

For uniqueness, if two solutions were different, take the smallest index r at which their coefficients differ. In (2), c is then nonzero, so their images under Ψ_m differ in degree $r+1$. They cannot both have image x^2.

For existence, start with $A = x$, for which $\Psi_m(A) = x^2 - mx^3$. Suppose the residual $\Psi_m(A) - x^2$ has already been made zero through degree r. Let c be its coefficient of x^{r+1}. Replacing A by $A - cx^r$ removes that coefficient without changing any earlier one, by (2). Continue for $r = 2, 3, \ldots$. Each coefficient of A eventually stops changing. Each coefficient of its defining equation depends on only finitely many coefficients of A, so the resulting series satisfies the equation in every degree.

More precisely, for $d \geq 2$, the coefficient $[x^d]\Psi_m(A)$ depends only on $a_0, \ldots, a_{d-1}$. The inner series T_A vanishes below degree two, so only outer indices $k \leq d/2$ contribute. Each coefficient of T_A through degree d depends only on coefficients of A through degree $d-1$. Thus later corrections to A cannot change a coefficient of the defining equation that has already been fixed. □

The proof uses no division by a coefficient. In particular it applies in residue rings even when the modulus is composite. Applying the lemma over $\mathbb{Z}$ defines A_m for every integer m. Its first terms are

$$A_m(x) = x + mx^2 + (2m^2 - m)x^3 + (5m^3 - 2m^2)x^4 + O(x^5). \tag{3}$$

For example, the coefficient of x^3 in the defining composition is $a_2(m) - m$, so $a_2(m) = m$.

Applying the same construction with an indeterminate parameter t over $\mathbb{Z}[t]$ shows that each $a_n(t)$ is an integer polynomial. Substitution $t = m$ then gives $a_n(m)$, by uniqueness. This observation will also allow a polynomial version of the divisibility theorem.

4 Parity

Theorem 4.1. *If m is odd, every $a_n(m)$ with $n \geq 1$ is odd. If m is even, then $a_1(m) = 1$ and every $a_n(m)$ with $n \geq 2$ is even.*

Proof. Suppose first that m is odd. Write P for the series obtained by reducing each coefficient of A_m modulo two. Equation (1) becomes

$$P(xP(x) + xP(x)^2) = x^2 \quad \text{in } \mathbb{F}_2[[x]], \qquad P = x + O(x^2). \tag{4}$$

The normalized solution is unique by Lemma 3.1. Consider the explicit series

$$H(x) = \frac{x}{1+x} = \sum_{n \geq 1} x^n \quad \text{in } \mathbb{F}_2[[x]].$$

The last equality is the geometric series identity from Section 2, with all signs equal modulo two. Direct calculation gives

$$xH(x) + xH(x)^2 = \frac{x^2(1+x) + x^3}{(1+x)^2} = \frac{x^2}{(1+x)^2} =: T(x).$$

Therefore

$$H(T(x)) = \frac{T(x)}{1+T(x)} = \frac{x^2}{(1+x)^2+x^2} = x^2,$$

since $(1+x)^2 + x^2 = 1$ in $\mathbb{F}_2[[x]]$. Thus H satisfies (4), and uniqueness gives $P = H$. Every nonconstant coefficient of H is one, so every $a_n(m)$ with $n \geq 1$ is odd.

If m is even, the reduced equation is $P(xP(x)) = x^2$. The series $P = x$ satisfies it and is the unique normalized solution by Lemma 3.1. Its coefficients after the first are zero, which proves the assertion for even m. □

Taking $m = 3$ proves the coefficient-parity conjecture in OEIS A396843.

5 An exact formula modulo m^2

The parity proof identifies a solution after reduction modulo two. We now use the same uniqueness principle modulo the parameter m, and retain one further level of information modulo m^2.

Theorem 5.1. *For every integer $m \geq 2$,*

$$A_m(x) \equiv x + m\sum_{j\geq 0}(-1)^j x^{2^j+1} \pmod{m^2}. \tag{5}$$

Equivalently, for $n \geq 2$,

$$a_n(m) \equiv \begin{cases} m(-1)^j \pmod{m^2}, & n = 2^j+1 \text{ for some } j \geq 0, \\ 0 \pmod{m^2}, & \text{otherwise.} \end{cases} \tag{6}$$

Proof. Reduce (1) modulo m. The resulting equation is $P(xP) = x^2$, whose unique normalized solution is $P = x$. Hence $A_m \equiv x \pmod{m}$, and we can write

$$A_m = x + mC, \qquad C \in x^2\mathbb{Z}[[x]].$$

The notation $C \in x^2\mathbb{Z}[[x]]$ means that C has integer coefficients and no terms of degree below two. Working modulo m^2 gives

$$T := xA_m - mxA_m^2 \equiv x^2 + m(xC - x^3).$$

Since $T \equiv x^2 \pmod{m}$, we also have $C(T) \equiv C(x^2) \pmod{m}$. Consequently,

$$A_m(T) = T + mC(T) \equiv x^2 + m(xC - x^3 + C(x^2)) \pmod{m^2}.$$

Equation (1) implies

$$xC(x) + C(x^2) \equiv x^3 \pmod{m}. \tag{7}$$

Here cancellation of m is ordinary integer divisibility: if m^2 divides md, then m divides d. It is applied separately to each coefficient.

Set

$$S(x) = \sum_{j\geq 0}(-1)^j x^{2^j+1} = x^2 - x^3 + x^5 - x^9 + x^{17} - \cdots.$$

The exponents increase, so this is a well-defined formal series. Its successive terms give

$$xS(x) = x^3 - x^4 + x^6 - x^{10} + \cdots, \qquad S(x^2) = x^4 - x^6 + x^{10} - \cdots.$$

Thus $xS + S(x^2) = x^3$ exactly.

The solution of (7) with no terms below degree two is unique over $\mathbb{Z}/m\mathbb{Z}$. Indeed, if a nonzero difference D began in degree $d \geq 2$, then xD would begin in degree $d+1$, while $D(x^2)$ would begin in degree $2d > d+1$. Its first term could not cancel in $xD + D(x^2)$. Therefore $C \equiv S \pmod{m}$, and $A_m = x + mC$ yields (5). □

6 Divisibility from binary digits

6.1 The binary digit inequality

Every nonnegative integer r has a unique binary expansion as a sum of distinct powers of two. Repeated division by two gives the binary digits as successive remainders, which also makes the expansion unique. Define $s_2(r)$ to be the number of powers in that sum, and set $s_2(0) = 0$. For example,

$$13 = 8 + 4 + 1 = (1101)_2, \qquad s_2(13) = 3.$$

Two properties will be used:

$$s_2(2r) = s_2(r), \qquad s_2(r_1 + \cdots + r_k) \le \sum_{i=1}^{k} s_2(r_i) \quad (r_i \ge 0). \tag{8}$$

Multiplication by two shifts every binary digit one place, proving the first identity. For the inequality, expand each summand into powers of two. Whenever a power occurs twice, replace the two copies by one copy of the next power. Each replacement preserves the value and decreases the number of summands. The process terminates at the binary expansion of the total, which proves the inequality.

6.2 The divisibility theorem

Let $B_m = x + \sum_{n\ge2} b_n(m)x^n$ be the compositional inverse of A_m, supplied by Lemma 2.1, and set $b_1(m) = 1$.

Theorem 6.1. *For every integer $m \ge 2$ and every $n \ge 1$,*

$$m^{s_2(n-1)} \mid a_n(m) \qquad \text{and} \qquad m^{s_2(n-1)} \mid b_n(m). \tag{9}$$

Proof. Fix $m \ge 2$. Remove the initial factor of x from the two inverse series:

$$A_m(x) = xF(x), \qquad B_m(x) = xG(x),$$

$$F(x) = \sum_{r\ge0} f_r x^r, \qquad G(x) = \sum_{r\ge0} g_r x^r, \qquad f_0 = g_0 = 1.$$

Then $f_r = a_{r+1}(m)$ and $g_r = b_{r+1}(m)$, so the shifted index r is the integer whose binary digits occur in the desired bound.

Apply B_m to (1). Using $B_m(A_m(x)) = x$ gives

$$xA_m - mxA_m^2 = B_m(x^2).$$

Hanna's entries record this identity for $m = 1, 2, 3, 4, 5$ [1, 2, 3, 4, 5]. We use it here for general m. Substitution of $A_m = xF$ and $B_m = xG$, followed by cancellation of the common factor x^2, yields

$$F(x) = G(x^2) + mxF(x)^2. \tag{10}$$

Cancellation here simply shifts every coefficient two places. For $r \ge 1$, comparison of coefficients gives

$$f_r = \begin{cases} g_{r/2}, & r \text{ even}, \\ 0, & r \text{ odd} \end{cases} + m \sum_{i+j=r-1} f_i f_j, \tag{11}$$

where i, j are nonnegative integers.

The other inverse identity, $A_m(B_m(x)) = x$, gives

$$1 = \sum_{j \geq 0} f_j x^j G(x)^{j+1}.$$

In degree $r \geq 1$, the $j = 0$ term contributes g_r and the $j = r$ term contributes f_r, because $G(0) = 1$. Thus

$$g_r = -f_r - \sum_{j=1}^{r-1} f_j [x^{r-j}] G(x)^{j+1}. \tag{12}$$

We prove by induction on r that

$$m^{s_2(r)} \mid f_r \quad \text{and} \quad m^{s_2(r)} \mid g_r. \tag{13}$$

The case $r = 0$ says that 1 divides $f_0 = g_0 = 1$. Suppose the assertions hold at every smaller index.

First consider f_r in (11). If r is even, then $r/2 < r$ and $s_2(r/2) = s_2(r)$, so the term $g_{r/2}$ has the required divisor. Each product $m f_i f_j$ uses $i, j < r$ and is divisible by $m^{1+s_2(i)+s_2(j)}$. The digit inequality (8) gives

$$1 + s_2(i) + s_2(j) \geq s_2(1 + i + j) = s_2(r).$$

Every term in (11) is therefore divisible by $m^{s_2(r)}$, and so is their sum f_r.

Now consider g_r in (12). The first term $-f_r$ has just been handled. For $1 \leq j < r$, expansion of G^{j+1} expresses each remaining summand as a sum of products

$$f_j g_{i_1} \cdots g_{i_{j+1}}, \qquad j + i_1 + \cdots + i_{j+1} = r, \qquad i_\ell \geq 0.$$

Each g index is at most $r - j < r$. By the induction hypothesis the product is divisible by

$$m^{s_2(j)+s_2(i_1)+\cdots+s_2(i_{j+1})},$$

and its exponent is at least $s_2(r)$ by (8). Taking sums and negatives preserves divisibility, proving the assertion for g_r. This completes the induction: at each stage we establish f_r first and then g_r.

Finally, replace r by $n - 1$ in (13). □

Remark 6.2. The same argument works before the parameter is assigned an integer value. Let t be an indeterminate and construct A_t and B_t over $\mathbb{Z}[t]$. All coefficient recurrences above use only sums, products, and negatives. The induction therefore proves that both $a_n(t)$ and $b_n(t)$ are multiples of $t^{s_2(n-1)}$ in $\mathbb{Z}[t]$. In other words, neither polynomial has a nonzero term of degree below $s_2(n-1)$ in the parameter t.

For illustration, the original parameter $m = 3$ gives the following bounds.

n	Binary expansion of $n-1$	$s_2(n-1)$	Guaranteed divisor of $a_n(3)$
2	1	1	3
8	111	3	27
9	1000	1	3
32	11111	5	243
64	111111	6	729

In general, $2^k - 1$ has k binary ones, so $m^k \mid a_{2^k}(m)$ for every $k \geq 0$. The table also shows why the guaranteed divisor need not grow with the index: a power of two has only one binary one.

6.3 Exactly where the smallest prime exponent occurs

For a prime p and a nonzero integer q, write $v_p(q)$ for the number of factors of p in q. Equivalently, it is the largest integer $e \geq 0$ such that $p^e \mid q$. For example, $v_3(45) = 2$. Set $v_p(0) = +\infty$. This notation lets us distinguish a lower bound from an exact exponent.

Corollary 6.3. *Let $m \geq 2$, let p be a prime dividing m, and put $e = v_p(m)$. Then*

$$v_p(a_n(m)) \geq e\, s_2(n-1) \qquad (n \geq 1).$$

For $n \geq 2$, equality with the smallest possible value e is characterized by

$$v_p(a_n(m)) = e \quad \Longleftrightarrow \quad n = 2^j + 1 \text{ for some } j \geq 0. \tag{14}$$

Proof. The lower bound follows immediately from Theorem 6.1. At $n = 2^j + 1$, Theorem 5.1 gives

$$a_n(m) = m((-1)^j + mq)$$

for some integer q. The factor in parentheses is 1 or -1 modulo p, so it is not divisible by p. Thus $a_n(m)$ has exactly e factors of p. At any other index $n \geq 2$, the positive integer $n-1$ is not a power of two and hence has at least two binary ones. The lower bound is then at least $2e > e$. □

7 Residue formulas for the original examples

For $m = 3$, Theorem 5.1 gives, for $n \geq 2$,

$$a_n(3) \equiv \begin{cases} 3 \pmod 9, & n = 4^k + 1 \quad (k \geq 0), \\ 6 \pmod 9, & n = 2 \cdot 4^k + 1 \quad (k \geq 0), \\ 0 \pmod 9, & \text{otherwise.} \end{cases} \tag{15}$$

These two exceptional cases correspond respectively to even and odd j in $n = 2^j + 1$. Equation (15) gives a full residue classification for A396843. The OEIS entry already records Danial Asaria's counterexample to a different proposed classification modulo nine [3]. His correction, dated 26 July 2026, uses $n = 5 = 4^1 + 1$: $a_5(3) = 975 \equiv 3 \pmod 9$, whereas that conjecture required residue one.

For $m = 4$, the corresponding formula is

$$a_n(4) \equiv \begin{cases} 4 \pmod{16}, & n = 4^k + 1 \quad (k \geq 0), \\ 12 \pmod{16}, & n = 2 \cdot 4^k + 1 \quad (k \geq 0), \\ 0 \pmod{16}, & \text{otherwise,} \end{cases} \qquad n \geq 2. \tag{16}$$

The residue conjecture in A396844 states the second case and divisibility by sixteen otherwise for $n > 8$ [4]. The first case in (16) is also necessary. For example, the entry's displayed term

$$a_{17}(4) = 113101684573952804 \equiv 4 \pmod{16}$$

has $17 = 4^2 + 1$. Formula (16) supplies the classification for all indices, including this case.

Computational reproducibility

The supplementary Python program `verify_hanna_congruences.py` constructs coefficients by successively imposing the defining equation, using the procedure in Lemma 3.1. It uses exact integer arithmetic and requires no third-party Python packages. Its default run checks the first 100 coefficients for each $m = 2, \ldots, 9$, including parity, the formula modulo m^2, the two binary divisibility bounds, and the minimum-exponent classification. It also checks the defining equation through degree 101, both inverse composition identities through degree 100, and the first twenty displayed OEIS terms for $m = 3, 4$. These finite checks provide reproducible examples; the proofs above do not depend on them.

Use of artificial intelligence and author responsibility

The mathematical arguments, Lean formalization, and computational code in this report were developed autonomously by ChatGPT (OpenAI) in response to research and revision prompts from the author. The author initiated the investigation and requested checks, extensions, and revisions; the author did not supply the proof strategy or intermediate mathematical arguments. ChatGPT also prepared the manuscript and performed the subsequent proof audit, citation checks, literature search, and revisions. The author has confirmed completion of human verification. Dev Goyal is the sole author and accepts full responsibility for the final article, including the correctness of its arguments, the accuracy of its references, the attribution of prior work, and any errors.

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